DI × BCG Circelligence + SAP

Competitive Product Analysis and Strategic Differentiation for Hilti Aftermarket Service (AMS)

Audience: Hilti Aftermarket Service stakeholders: business leadership, service operations, repair-centre teams, spare-parts and component recovery teams, circularity stakeholders, and digital and technology leadership.

Research cut-off: 5 November 2026.

Nature of the document: a rigorous product and competitive analysis. DI is a proposed product; no deployment, validated Hilti integration or measured result is claimed. Hilti-reported results are Hilti's and are not attributed to DI.

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Document purpose, scope and research cut-off

This report assesses whether DI, a proposed product described in the attached HTML file, has a credible and valuable role for Hilti Aftermarket Service (AMS) alongside capabilities Hilti already has through BCG Circelligence, the SAP ecosystem and its own operations. AMS always means **Hilti Aftermarket Service**: tool repair and maintenance, spare parts, service-centre operations, returned-tool assessment, refurbishment, component recovery and reuse, and end-of-life disposition.

Research cut-off: 5 November 2026. Sources were retrieved on that date or before. Later SAP, BCG or Hilti announcements are not reflected.

What the report is. A rigorous competitive product analysis. It is not a sales document, a plan or a proposal. It does not contain pilot, roadmap, pricing or go-to-market material.

What the report does not do. It does not claim that DI introduces repair, reuse, refurbishment or recycling to Hilti, that DI is deployed or integrated anywhere, that DI results have been measured, or that BCG or SAP cannot build comparable functionality. Hilti-reported results belong to Hilti.

Evidence classes (used throughout). **E1** confirmed deployed at Hilti. **E2** officially documented as part of Circelligence. **E3** officially documented as part of SAP. **E4** available through a third-party connector. **E5** possible through custom development. **E6** inferred. **E7** unknown. The phrase "not identified as a documented standard capability in the sources reviewed" means absence on the pages read, not proof of absence.

API verification levels (Section 7). **V1** SAP Help or SAP Learning text read. **V2** SAP Community, SAP KBA or SAP developer material. **V3** third-party listing or the DI HTML only. **V4** not verified.

Position codes (Section 8). A documented standard capability of Circelligence. B Hilti publicly confirmed using it. C SAP provides a relevant standard capability. D BCG + SAP could implement through configuration. E BCG + SAP could implement through custom development. F DI explicitly represents it in the HTML. G no clear DI differentiation. H uncertain or requires further evidence.

Status labels (Section 3B). GA generally available. Available. Announced. Limited release. Beta.

Citations. Source links appear inline next to the claims they support. Section 13 summarises which sources were read in full, which could only be seen as search excerpts, and which blocked automated access.

Executive summary

DI is a proposed domain decision layer for returned tools and recovered parts. In the sources reviewed, neither Circelligence nor SAP documents a standard feature for ranking recovery routes for individual items, matching recovered components to repair demand, or reconciling those decisions to outcomes. That is a publicly identifiable product gap. It is not a claim that BCG, SAP or Hilti cannot build such a layer, and Hilti may already hold equivalent logic that public sources do not show.

Figure 5. Competitive capability map

	Circelligence	SAP	DI (proposed)
Circularity scoring and material flows	Documented	Not a focus	Not claimed
Scenario modelling, initiative business cases	Documented	Not a focus	Case-level what-if
Repair application (quotation, order, confirmation)	Not a focus	Documented	Not claimed
Workflow, rules, human approval	Not a focus	Documented (generic)	Proposed
Agents, MCP, governance	Not a focus	Documented (generic)	Proposed
Integration, APIs, writeback	Not a focus	Documented (generic)	Proposed
Case-level route ranking and gates	Not a focus	Not identified	Proposed
Recovered-part-to-demand matching	Not a focus	Not identified	Proposed
Outcome reconciliation and learning	Not a focus	Not identified	Proposed

Documented in official or vendor sources · Not identified as a standard capability in sources reviewed · Proposed in the DI HTML, unvalidated

Key findings

- **Circelligence measures.** It is a strategic circularity assessment tool: quantitative and qualitative scores, material database, dashboards and a scenario player at company and value-chain level. Hilti confirms a portfolio score and database. Scenario use at Hilti is not confirmed.
- **Hilti already operates the activities.** It runs repair centres, reuses recovered parts through an automated, criteria-guided extraction process with individual part testing, assesses returned Fleet tools for pools, donation, parts reuse or recycling, and recycles through partners. Its published figures do not reconcile cleanly (Section 2) and are not DI results.
- **SAP is not just a system of record.** It provides applications (S/4HANA In-House Service, formerly In-House Repair), rules, workflow, Joule agents, an AI platform and integration. It documents the building blocks for every DI function except the domain content.
- **Generic agent, workflow, integration and governance features are weak DI differentiators.** SAP states its Business AI Platform is generally available, its agent gateway is generally available, and an Enterprise AI Gateway has been announced.
- **Third-party SAP offerings are narrow.** Cognite's writeback supports maintenance notifications and attachments only. WSO2's connector covers the Sales Order API only. Neither is a native SAP adapter.
- **DI's HTML is precise in what it represents.** It has route economics rather than a repair-cost model, eligibility checks rather than physical inspection or certification, one-off allocation rather than demand forecasting or scheduling, and illustrative API configurations rather than validated integrations. It has no standalone repair-versus-replace function.

> **Key competitive conclusion.** No DI wedge qualifies as proven Strong. Recovered-component-to-demand matching is the strongest conditional wedge, within a broader proposition of case-level aftermarket decision intelligence. Outcome reconciliation is how DI could show whether its decisions worked. The position rests on Hilti's unknown internal logic, real operational data and validation. The argument is a potential domain-specific build-versus-buy proposition, not proof that SAP or BCG cannot develop equivalent functionality.

The proposed DI wedge on one page

Element	Assessment
Wedge	Case-level aftermarket decision layer for returned tools and recovered components, with recovered-component-to-demand matching as the sharpest element and outcome reconciliation as the proof mechanism
Classification	Conditional. Not proven Strong
Positioning statement	DI is a proposed domain decision layer for returned tools and recovered parts. Using the facts held in SAP and other systems and the rules Hilti owns, it would rank eligible recovery routes and part allocations with stated economics and uncertainty, route the choice for approval into the system that executes it, and compare the outcome with the prediction. Circelligence measures circularity at portfolio level. SAP supplies the applications, rules, workflow and AI on which the transactions run, and on which a comparable layer could also be built
DI HTML modules	Returned assets, Asset intelligence, Route candidates, Six eligibility gates, Demand opportunity model, find_demand and optimize_pool, Pool optimizer, Approvals inbox, Predicted vs actual, Model versions, Audit and provenance, Connection Fabric
What SAP already supports	The underlying transactions: In-House Service confirmations with service-part and serial-number consumption, reservations and requisitions, stock and goods-movement APIs (graded **V1** to **V3**), plus rules, workflow, agents and adapters. Hilti's use of these is unknown
What Circelligence provides	Portfolio-level scoring, material database, dashboards and scenarios. Not case-level decisions
Standard functionality not identified	A component-to-demand matcher or allocation optimiser, route ranking with eligibility gates for recovered tools, and decision-to-outcome reconciliation, in the sources reviewed
What BCG and SAP could still build	Custom decision services on BTP using SAP APIs, rules, workflow and AI Core
What Hilti may already do	Automated criteria-guided part extraction and testing (2022). Matching to repair demand and route choice may exist in repair-centre or parts systems (**E7**)
Data required	Recovered inventory by part and location, repair-order and expected demand, compatibility, quality rules, new-part price and lead time, cost data, actual outcomes
Evidence that would show value	Share of recovered parts consumed, new-part purchases avoided, recovered-stock ageing and repair wait, predicted-versus-actual error, compared with historical practice
Why no moat is proven	The features are replicable; any advantage would come from reconciled outcome history and Hilti-tuned rules, which accrue only with use and may belong largely to Hilti

1. Circelligence: methodology, product and what it is not

Circelligence is a strategic circularity assessment and scenario tool, not an operational decision system for individual tools or parts. BCG's own product page describes it as "part evaluative, part circular economy framework" that gives an end-to-end assessment of a company's circularity and helps leaders identify initiatives. Everything it documents sits at the level of company, portfolio and value chain.

Three things kept separate

- *Methodology:* BCG's circular-economy assessment framework, developed with Hilti from 2020 and co-developed with Circle Economy (Hilti and TechTarget material).
- *Current product:* the software BCG describes today, offered as a standalone circular strategy solution running on SAP BTP, built by BCG X with SAP (BCG product page, 2026 copyright footer, undated).
- *Hilti's deployed implementation:* only what Hilti and SAP publicly state (Section 2).

Current product, as BCG describes it (E2)

Pillar on BCG's page	What the page says	Level of analysis	Evidence
Quantitative assessment	Baseline of circularity performance; pulls available data from SAP systems to assess the full value chain from raw materials to end of life; data can also be entered through Excel templates or the user interface	Company and product value chain	E2
Qualitative assessment	Survey questions gauge cultural and organisational readiness to embrace and execute a circular roadmap	Organisation	E2
Visibility and transparency	A dashboard of circularity scores for current performance and improvement areas	Company and portfolio	E2
Simulating improvement initiatives	A scenario player models how changes to processes and raw materials affect circularity; supports testing initiatives, developing and assessing business cases, and prioritising	Strategy and portfolio of initiatives	E2
Customisation	Can be tailored to a company's operating model and ways of measuring success	Configuration or consulting	E2

What the page does not say. It does not mention individual repair cases, returned-tool disposition, component eligibility, inventory, demand, approvals, execution or writeback to SAP. These are "not identified as a documented standard capability in the sources reviewed". The page describes business-case assessment only for improvement initiatives modelled in the scenario player, so strategic business cases are in scope and per-case operational economics are not described.

Business-model note. The page offers a demo request and presents BCG consulting experts alongside the tool, so the product comes with BCG methodology and consulting. What a Hilti licence includes is E7.

Source caution. The BCG page is marketing material and undated. Its Hilti claims are checked against Hilti's own reports in Section 2.

1A. The BCG Circelligence framework in detail

This subsection describes the framework as BCG and Circle Economy published it in November 2021, in "Circular Metrics for Business: Introduction to the CIRCelligence indicators framework" (BCG and Circle Economy, version 1.0, opened in full; the paper is licensed CC BY-SA 4.0). It is the most detailed public description of the methodology. The current software product (Section 1) and Hilti's deployment (Section 2) came later and may differ. Evidence class **E2** for the framework, not E1.

Provenance and maturity. BCG launched the framework in February 2020 and Circle Economy assisted in developing the methodology; BCG is its co-author. The paper describes the proprietary metric and tool as technically complete after several client projects. On Circle Economy's six-level Methodology Readiness Level scale it placed Circelligence at level 2, "tested within a limited set of applications", as of 2021; that rating predates later deployments, including Hilti's. The Ellen MacArthur Foundation's 2024 landscape report classifies it as a voluntary framework that is part evaluative and part circular-economy framework.

Structure: five value-cycle steps and two business functions.

Element	What it covers	Sub-indicators	Quantitative or qualitative
Input	Resources flowing into the company	6	Both
Product design	Design and material choices	4	Both
Production and sales	Production resource and material flows	10	Both
Business model and usage	Use phase and lifetime	6	Both
End of life	Treatment and recovery of products	8	Both
Business implementation	Strategy, steering and governance	12	Qualitative only
Collaboration and ecosystems	Partners across the value chain	9	Qualitative only

For each value-cycle step the framework looks at inflow (circular versus linear resources entering), slow flow (lengthening versus shortening the duration of use), outflow (circular versus linear resources leaving) and qualitative aspects. It reads material mass flow and material value flow, and a qualitative layer for steering and ecosystem engagement.

Scores and indicators. The framework has more than 60 indicators in three kinds: a headline indicator (percent circularity, the average of all performance indicators), performance indicators for each value-cycle step, and process indicators for the two business functions. Quantitative results run from 0% to 100% and can be computed on a mass basis and a value basis. Qualitative results are graded A to F for how far circular thinking is embedded in processes and governance. The paper says this pairing separates coincidental from intentional circularity: a high quantitative score with a weak qualitative grade is "coincidental and at risk".

Data and effort. The calculator collects data through a customisable web-based survey of about 50 quantitative questions (with up to 2,500 data points) and about 100 qualitative questions. It is set up for nearly 100 input materials and can analyse end-of-life effects across up to 10 sales markets. Data from supply-chain partners may be needed; where gaps remain, BCG suggests sector averages or proxies. A reduced scope is possible: excluding the value analysis, analysing only some steps, or leaving out the qualitative part. The paper states that all data stays in company ownership.

Three-step approach.

Step	What it does	Outputs
1. Create transparency	Gather quantitative and qualitative data and calculate the circularity baseline across the value cycle	Scores and grades per step and function; root causes; a clear view of competitive advantages and gaps
2. Develop an actionable roadmap	Simulate scenarios for required activities and set an ambition level for what can reasonably be achieved in a time frame; BCG gives recommendations	Ambition level, leverage points, quick wins
3. Implement key initiatives	Identify KPIs to track progress; use for internal and external communication, from board level to sustainability reporting	KPIs, progress monitoring, stakeholder reporting

Lenses and dashboard. The calculator can focus on product, product component, packaging, division or company-wide performance. A dashboard shows the whole value chain and each segment, with scores and grades on mass and value bases, ambition levels and key initiatives. The paper's example traces a division's high score to a supplier of recycled PET that is price-competitive with virgin PET, which suggests other divisions could switch.

Delivery model. The first application is always paired with a BCG project and on-the-ground consultants. Later use can be done by the client with a self-assessment tool, which in 2021 was described as a free online tool, with the same flexibility to use the full or a reduced scope. BCG also offers cost-benefit assessments to prioritise initiatives with the greatest economic impact and fewest trade-offs. The paper positions the outcomes for executives and CSR or innovation managers, and as consistent output for investors and external reporting.

How the framework evolved in the sources reviewed.

Source	Date	What it adds
BCG and Circle Economy paper	Nov 2021	Survey-based calculator, five steps, 60+ indicators, A to F grades, mass and value scores
Hilti, BCG and SAP release	22 Nov 2023	Web-app interface and SAP BTP deployment; automated database updates; scenario assessments described as groundwork, not as deployed
BCG product page	Undated, 2026 footer	Five pillars: quantitative (SAP data or Excel), qualitative survey, visibility dashboard, scenario player, customisation

Where the framework stops and DI begins. The framework's documented solutions work at company, division, product and component level and answer how circular an organisation is and which initiatives would raise that. DI's HTML works on individual returned items and answers what to do with each one. Based on the DI HTML, DI does not represent a circularity baseline, a qualitative readiness assessment, 0 to 100% or A to F benchmark scores, or a strategic ambition and initiative roadmap. DI's Sustainability view reports CO2e avoided by route, which is a different metric. This is a statement of scope, not a development plan. Neither product is shown to be better than the other; they sit at different levels.

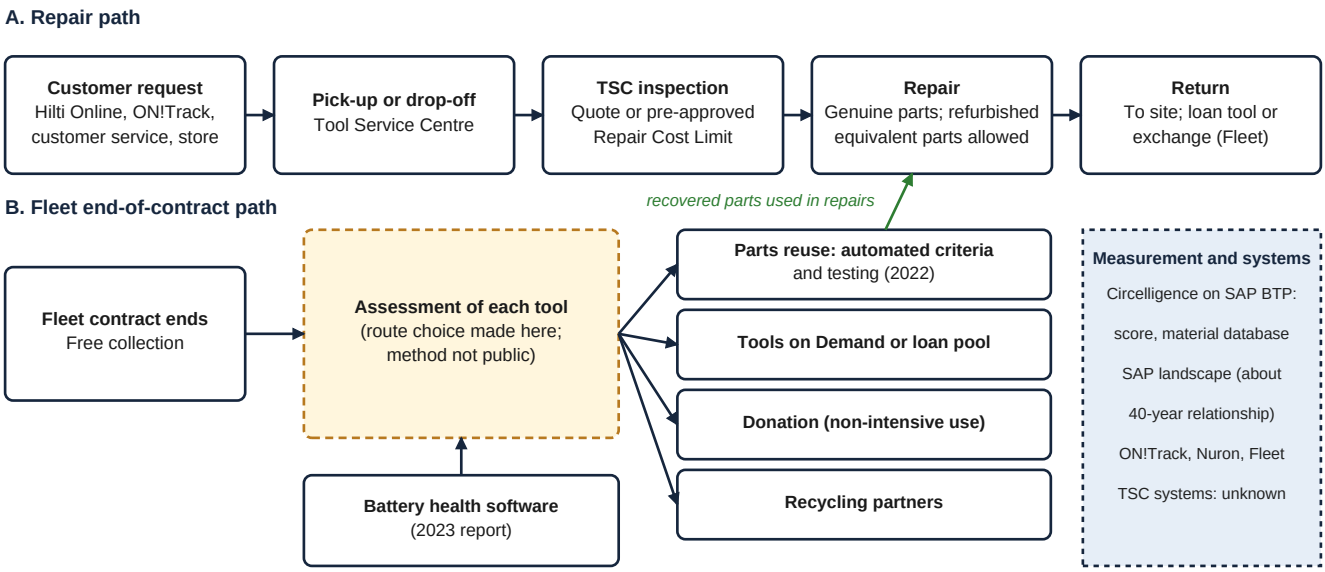
Framework solution	Level	DI in the HTML
Circularity baseline, mass and value	Company, value chain	No equivalent. Sustainability view reports CO2e estimates versus measured values by route
Qualitative readiness grades A to F	Organisation	No equivalent
Lenses by product, component, division	Portfolio	Asset and component intelligence at item level, a different purpose
Scenario simulation and ambition setting	Strategy	What-if on a single case only
KPIs and progress reporting	Company	Route mix, Predicted vs actual and Overrides at decision level
Cost-benefit prioritisation of initiatives	Portfolio of initiatives	Per-case route economics
Approval, handoff and execution of an item-level decision	Not in scope of the framework	Approvals inbox, handoff and reconciliation, proposed

Caution on third-party summaries. An outside write-up of Circelligence versus a proprietary workbench was compared against this primary source. Its factual points on the framework (five value-cycle steps, more than 60 indicators, mass and value scores, A to F grades) match the BCG and Circle Economy paper. Its claims that the proposed product is stronger than Circelligence in operational execution are not supported by evidence in this research and are not adopted here.

2. Hilti's documented use of Circelligence

Public sources confirm that Hilti uses Circelligence to score and steer circularity across its product portfolio. They do not attribute any case-level repair, disposition or part-matching decision to it. The repair-centre reuse process is described as Hilti's own operation, with Circelligence credited for measurement and strategy.

Figure 2. Hilti aftermarket service context (documented activities and unknowns)



Solid boxes: documented by Hilti or SAP. Dashed amber: analytical interpretation. Where DI would sit is shown in Figures 3 and 4.

Topic	What public sources say	Source	Class
Origin	Hilti says it was the first company to use and develop the method with BCG, co-developed with Circle Economy; partnership began in 2020	Hilti 2021 report, TechTarget	E1
Circularity score	Hilti can calculate a qualitative and quantitative circularity score	SAP, 22 Nov 2023	E1
Portfolio assessment	Entire product portfolio evaluated yearly; results published in the sustainability report and used as a basis for Scope 3 CO2 goals and Science Based Targets initiatives	Hilti Group release	E1
Material database and flows	Automated updates to BCG's materials database on SAP BTP; tracking from input through design, production, business model and usage, and end of life, in a web interface. Hilti mapped about 250,000 tons of material into groups such as steel, aluminium, plastics and copper	SAP, TechTarget	E1
SAP data and dashboard	Sustainability data from SAP systems brought into Circelligence with a dashboard-style interface (trade press account, 5 Feb 2024)	TechTarget	E1 (secondary)
Qualitative score	The 2023 Sustainability Report cites an improved qualitative score in product design, business model and end of life	Hilti 2023 report	E1
Scenario player	In November 2023 BCG said the latest evolutions "lay the groundwork" for integrating interactive scenario assessments and future automation, so scenario use at Hilti is not confirmed	SAP	E7
Part-type trade-offs	Hilti's head of circular economy says Hilti analyses which parts make most financial sense to reuse versus which save most CO2, finding electronics more effective than steel, and focuses on the highest-value parts	TechTarget	E1 (secondary; part-type level)
Remanufacturing	Remanufacturing at 70 repair centres: electronics, motors, rotors and other parts extracted from returned equipment and reused after extensive testing; non-reusable parts recycled by qualified partners. SAP says the database transparency "helped define" these initiatives	SAP	E1 as a supplier claim
Reuse traceability in SAP	Hilti's head of circular economy says S/4HANA tracks products once from sourcing to sale, so reused items had no identity in it. Hilti needed "a lot of additions to the SAP landscape" so that reused electronics are kept separate from new ones for quality and linked to their parent tool, while stock and forecasting count them together	TechTarget, 5 Feb 2024	E1 (secondary)

Topic	What public sources say	Source	Class
Reuse assessment equipment	The equipment that lets a technician assess whether something is reusable involves algorithms, testing machinery and a physical setup	TechTarget, 5 Feb 2024	**E1** (secondary)
Repair-centre process	An automated process, started in 2021, guides technicians end to end and defines when each part is extracted by prescribed criteria; each reused part is tested individually with the same guarantee as new; data is collected to improve the process	Hilti 2022 report	**E1**, attributed to Hilti, not to Circelligence

The 2022 figures, checked. BCG states that Circelligence insights led to "the recovery of 120,000 spare parts in 2022 and saving 90 tons" (BCG page). Hilti's own sources give a different series:

Reported figure	Source and reporting basis
10,000 to 100,000 parts a year within two years; about 50 tons a year	2021 report
100,000 parts a year, grown from 10,000 within three years; 90 tons a year	2022 report
"Up to 120,000 per year", about 90 tons a year	Hilti Corporation circularity page, undated
160,000 refurbished parts, 16 times in three years	Hilti US sustainability page, reporting year not stated
190,000 parts reused in 2024; about 90 tons	Hilti NZ circularity page

Reading. The 120,000 figure on BCG's page is not matched by Hilti's 2022 company report, which gives 100,000 for the same period; 120,000 appears on a Hilti page as "up to" and undated. The "90 tons" figure is reported unchanged next to 100,000, 120,000 and 190,000 parts, so it cannot be a consistent per-year measure and should be treated as indicative (the 2021 report gave about 50 tons for 100,000 parts). The 160,000 figure on Hilti's US page has no stated year; the same page cites 2023 for its CO2 progress, so it probably reflects 2023, which is not confirmed. The page carrying "up to 120,000" presents 2020 collection data (1.3 million items collected in 2020), which suggests that wording is older than Hilti's 2022 report and may predate rather than follow it. These are Hilti-reported results of Hilti operations, and they are not attributable to DI. Whether Circelligence caused them is a BCG claim that public sources do not test.

What this establishes for DI. Circelligence at Hilti is confirmed as a measurement, database and strategy layer. The operational decisions on the repair-centre floor are described as Hilti's own automated process. Hilti has also extended its SAP landscape to track reused items, which shows internal engineering around reuse that public pages describe only briefly. Whether Hilti has private or customised Circelligence functionality beyond these descriptions is E7.

3. SAP BTP and Integration Suite

SAP is not only a database or system of record. It provides, as documented products, applications, rules, workflow, AI, agents and integration. The building blocks cover every DI function except the aftermarket-specific decision content, and SAP provides one prebuilt aftermarket application, S/4HANA Service with In-House Service (called In-House Repair before release 2025). SAP's building blocks make a custom decision layer on BTP technically credible. What the sources reviewed do not identify is a prebuilt route-ranking, component-eligibility or recovered-part-matching application. Licensing and deployment at Hilti are **E7** for every item below.

Relevant SAP portfolio, as documented

- *Integration Suite* (**E3**): Cloud Integration adapters including OData **V2** and **V4**, SOAP, IDoc, RFC, SFTP, AMQP and Kafka; Open Connectors for non-SAP applications (the Feature Scope Description cites over 150, an SAP user-group deck cites 250+); API Management; Event Mesh included in the standard and premium editions; Advanced Event Mesh as a separate capability with direct consumption of S/4HANA events. Sources: SAP Feature Scope Description, 13 Aug 2026, SAP Integration Suite overview, Mar 2025, SAP user-group deck, Feb 2026.
- *Business AI platform* (**E3**): Joule Studio Agent Builder, described as generally available since January 2026 in an SAP Community post that blocked automated access, with MCP support, document grounding through AI Core, and human oversight "by default for critical decisions" (SAP Community). Agents run on SAP AI Core with built-in metering, tracing and security; SAP Build Process Automation provides workflows and business rules as agent tools; SAP uses MCP internally and supports A2A and MCP interoperability (SAP architecture guide). SAP AI Agent Hub is described as a control pane for agents, models and MCP servers with governance assessments (SAP, Q2 2026 highlights). At TechEd 2026 (28 Oct) SAP said its agent gateway is generally available and announced SAP Enterprise AI Gateway, built on Integration Suite, for rate limiting, routing, protocol translation and advanced access control; general availability of the Enterprise AI Gateway is not stated in the sources read. Status detail is in Section 3B.
- *Service application* (**E3**): S/4HANA Service with In-House Service (renamed from In-House Repair with release 2025; see Section 3A), covering in-house service objects, prechecks, service quotations, service orders, service confirmations, spare-part consumption, billing and returns logistics (SAP Learning, SAP Community). Whether Hilti runs it is E7.
- *Not reviewed in detail in this research:* SAP Datasphere, SAP HANA Cloud, SAP Build Apps and SAP Analytics Cloud. They are generic data, analytics and app services and are treated as **E7** for any specific DI-like function.

Capability classification

DI-like function	Standard SAP (documented)	Configure	Custom build	Extra SAP product or licence	BCG methodology or consulting	Not established in sources reviewed
Rule-based gates	Business rules in SAP Build Process Automation (E3)	Gate rules and thresholds	Hilti's reuse and eligibility content	Build Process Automation is a BTP service; Hilti licence E7	None	A prebuilt component-eligibility rule set for tools
AI-assisted recommendation	Generative AI hub and AI Core; Joule agents grounded in documents and data (E3)	Agent instructions and grounding	Prediction models for repair cost, success, price	AI Core and Joule Studio usage terms E7	None	A prebuilt route-recommendation model for returned tools
Workflow and human approval	Build Process Automation workflows; human-in-the-loop in agents (E3)	Approval steps and roles	Delegation and spend-cap logic	As above	None	A prebuilt decision-approval workflow for disposition
Process execution and writeback	S/4HANA OData APIs and Integration Suite adapters (E3)	Adapter flows	Mapping of decision to SAP objects	Integration Suite (E7 licence)	None	A prebuilt decision-to-SAP handoff for disposition
Data reconciliation	Message monitoring and error handling in Integration Suite (general product feature)	Monitoring set-up	Business reconciliation of predicted to actual	None	None	Prebuilt predicted-versus-actual reconciliation
Decision auditing	AI Core tracing; AI Agent Hub governance and runtime verification; agent gateway (GA); Enterprise AI Gateway observability (announced)	Logging	Decision records with evidence snapshots	AI Agent Hub, agent gateway, Enterprise AI Gateway (licence E7)	None	A prebuilt decision audit model for disposition
Agent and tool orchestration	Joule Studio, MCP, A2A, Integration Suite-managed MCP servers (E3)	Agent and tool wiring	Domain tool catalogue	As above	None	SAP-built tools for repair and recovery decisions
Optimisation across routes	No standard route-allocation optimiser identified	n/a	Custom optimiser	Other SAP planning products not reviewed (E7)	None	Component-to-demand allocation optimiser
Analytics	Datasphere, HANA Cloud, SAC (not reviewed)	Dashboards	Domain analytics	E7	Circelligence dashboards for circularity	Case-level recovery analytics
In-house service execution	S/4HANA In-House Service (E3)	Process configuration	Extensions	S/4HANA Service licence (E7)	None	Route ranking beyond the precheck decision on follow-up steps for in-house service objects

Reading. For every row, SAP can host a custom build, and SAP does not need DI to execute, approve or integrate. The open question is whether Hilti wants to build the domain decision content (gates, economics, matching, reconciliation) from building blocks or adopt a product that already carries it. That is a build-versus-buy argument, not a capability gap.

3A. SAP In-House Service (formerly In-House Repair): current scope and overlap with DI

Verification note. Of the four SAP pages supplied for this topic, three on help.sap.com disallow automated access and could not be read. The SAP Learning page was read in full (Planning and Performing In-House Services). Other facts come from SAP Learning, SAP Help and SAP Community excerpts returned by search, and one member-authored post, each graded below. Hilti's use of this application is E7.

Terminology

Older term	Current term	Source and status
In-House Repair	In-House Service, from release 2025 in S/4HANA and S/4HANA Cloud Private Edition; SAP Learning notes older releases used "in-house repair"	**V1** SAP Learning, configuration lesson
Repair object	In-house service object	**V1**, same sources
Solution process	In-House Service (7GB) for S/4HANA and Private Edition; the SAP Learning material read for Public Edition still names scope item In-House Repair (3XK), and whether Public Edition has been renamed was not verified	**V1** for 7GB; **V2** for 3XK
Apps Perform Prechecks, Plan Repairs, Perform Repairs, Process Repair Quotations, Prepare for Billing, Manage Repair Quotations, Orders and Confirmations	Deprecated as of S/4HANA 2023 FPS02; successors include Manage In-House Service Objects and Release for Billing	**V1** SAP Learning
Repair-specific transaction types (REPA, REPC, REPO, REPQ, REPT)	In-house service IHRP (item category IHRI), service order SRVO and service confirmation SRVC. A new process version, "In-House Service 2.0", was introduced with S/4HANA 2023 FPS02, and the older transaction types have been deprecated since then	**V1** SAP Learning, configuration lesson. A member-authored SAP Community post places In-House Service 2.0 at 2025 FPS02; the SAP Learning text contradicts that and is followed here (the post could not be opened)

Process and objects in current SAP documentation (V1 unless noted)

Step or area	Current SAP capability
Intake	In-house service created from a service quotation or service order, from the Manage In-House Services app, or from a returns order follow-up activity; AI-assisted initiation extracts in-house service objects from scanned or photographed paper documents
Precheck and follow-up	A precheck on each in-house service object decides the follow-up steps, for example creating a service order or managing customer returns. Before 2023 FPS02 user statuses such as Precheck: Decision Pending tracked this; since then status is shown mainly through the system statuses of the service order or quotation, and the process-step overview shows steps such as Precheck, Repair and Completed (**V1**, SAP Learning configuration lesson)
Service quotation	Quotation with estimated costs to the customer; requote when additional items are needed
Service orders and confirmations	Planning in service orders; actual consumed resources and parts confirmed in service confirmations. SOAP and OData **V4** APIs exist for service orders (**V1**, SAP Learning); a service-order SOAP API gained warranty information, In Process status and change indicators in release 2025
Spare parts	Reservations for stock service parts and purchase requisitions for externally procured parts (**V2**, SAP Learning); serial numbers of consumed serialised parts (**V2**, Public Edition 2302 post); lists of serial numbers on one service object in 2025 (**V2**, SAP Community)
Returns and logistics	Returns order, posting to customer stock, outbound delivery, intracompany and intercompany routing to repair centres, splitting and merging service objects (**V2**, SAP Help, SAP Community)
Advanced execution	Combines maintenance orders with service quotation and service order, and billing
AI	Joule search and follow-up creation for in-house service objects (**V2**, SAP Community)

Step or area	Current SAP capability
APIs	OData **V4** APIs for service order, service confirmation and service quotation in 2025, with the **V2** service order and confirmation APIs deprecated (**V2**, SAP Learning); an entry "OData API: In-House Service" appears in SAP Help what's-new navigation for 2025 FPS02 (**V3**, name only, SAP Help excerpt)

Overlap with DI's proposed repair and returned-tool workflow

DI workflow element (HTML)	SAP In-House Service	Overlap
Returned record and evidence (Returned assets, Asset intelligence)	Returns order and in-house service object hold the return and its data	Data source for DI, not a duplicate function
Inspect and grade	Precheck and diagnosis steps record the result; DI does not inspect	DI would read the result
Decide the next step (Route candidates, gates, ranking)	Precheck is a human decision on follow-up steps for each object; no ranking, eligibility gates or economic comparison across routes is described	Partial: the same decision point, different content. SAP provides the container, DI proposes the ranked content
Customer quotation	Service quotation and requote	SAP only; DI has no quotation
Repair execution and confirmation	Service orders, confirmations, maintenance orders in advanced execution	SAP only; DI hands instructions to EAM or MRO systems
Spare-part consumption and stock	Confirmations, reservations, requisitions, serial numbers	SAP holds the transactions; DI reads and, in its illustrative configuration, reserves
Returns and logistics	Returns order, deliveries, intercompany routing	SAP holds the transactions; DI models movement cost
Routes beyond repair (harvest, parts allocation, resale, material recovery, pool allocation)	Not described as in-house service functions	DI only as designed; not identified in SAP
Predicted versus actual	Confirmations capture actuals; no comparison with a prediction is described	DI only as designed

Reading. The overlap with SAP's current application is concentrated in the repair-workflow container and its transactions, and DI's HTML does not try to replace those. DI and the application meet at the precheck decision. If Hilti runs In-House Service, a DI recommendation could inform that decision (**E6**, inference; no such integration exists). If Hilti does not, the application is irrelevant. SAP is known at Hilti as its ERP supplier of about 40 years and the host of Circelligence, and Hilti has described building additions to its SAP landscape to track reused items (TechTarget, 5 Feb 2024); its role in Hilti's service centres is otherwise E7.

3B. SAP AI and gateway status as of 5 November 2026

The requested SAP News Center page dated November 2026 returned an error. Its content was read from the SAP Africa News Center version dated 2 Nov 2026, and the TechEd keynote summary dated 28 Oct 2026 was read in full. Statements marked "reported" are executive quotes in those pages.

Item	Status in official SAP material	Label	Sources
Joule Studio Agent Builder (original)	Described as generally available since January 2026, with MCP, document grounding and human oversight for critical decisions. This rests on an SAP Community post that blocked automated access and was not opened	**GA** (per community post, not opened)	SAP Community
New Joule Studio	SAP's Q2 2026 highlights (20 Jul 2026) list the new Joule Studio under SAP Early Adopter Care, with a trial still to open and a migration path from the original	**Limited release** as of July 2026; SAP's CTO is reported on 2 Nov as calling Joule Studio generally available	SAP Q2 highlights
SAP Business AI Platform, Knowledge Graph, Business Data Cloud, Joule Studio, Joule Work, agentic orchestration and integration	SAP's CTO is reported saying the platform "is now generally available" with all its capabilities	**GA** (reported, 2 Nov 2026)	SAP Africa News Center

Item	Status in official SAP material	Label	Sources
SAP AI Agent Hub	Q2 2026 highlights (20 Jul) list enhancements such as a verification badge and say runtime observability and governance are "looking ahead". The TechEd keynote (28 Oct) describes runtime verification enforcement: agents not verified in the Hub cannot be invoked	**Available**; separate GA date not stated	SAP TechEd, 28 Oct 2026, Q2 highlights
Agent gateway	"Generally available today", connecting agents and endpoints across landscape boundaries	**GA**	TechEd 28 Oct 2026
SAP Enterprise AI Gateway	"Announcing"; built on Integration Suite as an overlay to the agent gateway for rate limiting, routing, protocol translation and advanced access control; described as delivering policy enforcement, observability and runtime controls. No availability date stated	**Announced**	TechEd 28 Oct 2026
AI Governance Assistant	Released in November to Early Adopter Care program customers	**Limited release**	TechEd 28 Oct 2026
Business Data Cloud Ontology Manager	**Available** as beta	**Beta**	TechEd 28 Oct 2026
SAP-RPT-2.0 tabular model	**Announced**; anomaly and outlier detection and embeddings described as expanding	**Announced**	SAP Africa News Center
Joule assistants and agents	40 assistants and 300 agents between SAP and partners announced; whether any covers service repair or recovered-part decisions was not found	**Announced**; domain coverage unknown	TechEd 28 Oct 2026

Correction to the earlier version of the research. The earlier text described auditing through an Enterprise AI Gateway as planned. The current status is: agent gateway generally available, Enterprise AI Gateway announced without a stated availability date, AI Agent Hub available with runtime verification. SAP's own framing is that its agents are embedded in deterministic application workflows, run on the same data and rules, and leave the same audit trail as the applications.

Impact on DI. The conclusion stands and is slightly reinforced. Generic agent building, MCP, orchestration, human oversight and agent governance are current or announced SAP capabilities, so they are weak differentiators for DI. SAP's AI claims are about its own applications and SAP-governed agents. None of the pages read describes an SAP agent for route choice or component matching for recovered tools.

4. SAP adapters and connectors: classification

The supplied references mix different kinds of integration offering. The Cognite writeback and the WSO2 connector are third-party and are not SAP-native adapters.

Class	Examples in this assessment	What it is	Evidence
SAP-native adapters	Cloud Integration adapters for OData **V2** and **V4**, SOAP, IDoc, RFC, SFTP, AMQP, Kafka, HTTP and mail; Advanced Event Mesh adapter	Connectors inside SAP Integration Suite, configured as communication channels in integration flows	**E3**: SAP Feature Scope Description, SAP Community adapter guide
SAP APIs	S/4HANA OData APIs such as Service Order, Maintenance Notification, Material Document, Product; business events via Event Mesh	Interfaces of SAP applications that adapters or other clients call	**E3** for Service Order; others listed in Section 7 with verification status
SAP product for third-party apps	SAP Open Connectors (over 150 pre-built connectors to non-SAP applications)	An SAP service in Integration Suite, but it connects non-SAP applications	**E3**
Third-party SAP connectors	WSO2 SAP Sales Order connector	Open-source client for one SAP API, built on the WSO2 and Ballerina platform	**E4**: WSO2
External integration-platform writeback	Cognite Data Fusion SAP writeback	A Cognite service that writes CDF data into SAP maintenance notifications and attachments	**E4**: Cognite
Custom integration code	OData clients generated with the SAP Cloud SDK; partner-built adapters through the adapter framework	Code or adapters built per project	**E3** for the SDK (it supports OData services listed for S/4HANA Cloud per SAP Cloud SDK notes); **E5** for project code

What the DI HTML says about SAP. The ERP source in the DI HTML is illustrated with SAP S/4HANA and shows read calls to API_MATERIAL_STOCK_SRV (A_MatlStkInAcctMod), API_PURCHASEORDER_PROCESS_SRV and API_PRODUCT_SRV, and write calls to API_RESERVATION_DOCUMENT_SRV and API_MATERIAL_DOCUMENT_SRV, with OAuth 2.0 and mutual TLS. These are illustrative configurations in a proposed product and are not validated against any Hilti landscape.

Which adapters could connect DI's proposed source systems to a Hilti SAP landscape (E6, inference)

DI source type	SAP-side pattern	Notes
ERP (stock, cost, purchase baseline)	Direct OData **V2** or **V4** calls, or Integration Suite OData adapter	Matches the DI HTML configuration; API availability by edition must be confirmed
EAM and MRO or workshop	S/4HANA Service or Plant Maintenance APIs through the OData adapter; IDoc or RFC where APIs are missing	In-house service objects (formerly repair objects) and service orders are SAP application objects. An entry titled "OData API: In-House Service" appears in SAP Help what's-new navigation for 2025, but its name, scope and operations were not read, so API coverage of in-house service objects is unverified
WMS and stock movements	Material document and stock APIs; Event Mesh for movement events	Depends on whether Hilti uses S/4HANA warehouse functions or another WMS
Quality (QMS, inspection)	Inspection lot and quality notification APIs (SAP_COM_0318 scenario for inspection lots)	API availability not verified in Hilti's edition
PLM and BOM	Product and BOM APIs, or a non-SAP PLM through Open Connectors or custom adapters	Hilti's PLM landscape is **E7**
Events	Event Mesh or Advanced Event Mesh consuming S/4HANA business events	Event catalogue for service and stock objects not verified
File and legacy	SFTP, IDoc, RFC adapters	Standard Integration Suite adapters

Standard versus custom. SAP provides the transport and the APIs. Mapping DI's decision, reservation and instruction objects to SAP documents, error handling, idempotency and access security are integration-project work, whoever builds it.

Edition caution. Public Edition, Private Edition and on-premise differ in available APIs (for example the **V2** service-order API is deprecated from release 2025 and replaced by an OData **V4** API in Private Edition, per SAP Learning). Hilti's edition and enabled communication scenarios are E7.

5. Cognite Data Fusion SAP writeback: exact scope

Cognite's SAP writeback is a third-party, generic writeback pipeline from Cognite Data Fusion (CDF) with two supported SAP entities, maintenance notifications and attachments. It is not a native SAP adapter (class **E4**), and it says nothing about service orders, in-house repair or in-house service, stock or component recovery. Sources: SAP writeback service and setup guide, read in full on 5 Nov 2026; neither page shows a date.

Question	What the documentation states
What it is	A service that sends data from CDF to SAP S/4HANA through configured SAP-standard OData V2.0 endpoints, to create and update SAP records including attachments
Supported SAP entities	Maintenance notifications (API OP_API_MAINTNOTIFICATION) and attachments (API OP_API_CV_ATTACHMENT_SRV_0001), and no others are listed
Maintenance notification operations	Create (example with NotificationText, NotificationType, MaintPriority, long text) and update (the SAP object ID must be in the payload). Reading notifications from SAP is not described on these pages; the service is a writeback
Attachments	Attach a CDF file (CogniteFile or legacy file) to an SAP object, with the SAP notification ID as sourceId; the service fetches file content from CDF
SAP versions	S/4HANA on-premise 2021 FPS01 and later; S/4HANA Cloud. The page does not distinguish Public from Private Cloud edition
Service orders	Not listed as supported
In-house repair or in-house service objects	Not listed as supported
Material and inventory records	Not listed as supported
Component recovery transactions	Not listed as supported
Using other entities	Not stated. The setup describes an "endpoint" as a configuration to an SAP OData endpoint and its entity, which suggests a generic mechanism, but the supported-entity statement limits what Cognite documents. Whether other entities work is **E7** and would need Cognite confirmation
Prerequisites	Cognite API and apps registered in Microsoft Entra ID; CDF groups and SAP writeback capabilities; SAP destination endpoints active and reachable from outside the intranet; an SAP user authorised for OData services
Mechanics	Create an SAP instance (host), optional schema mapping in Cognite's mapping language, an SAP endpoint, an optional connectivity check, then asynchronous requests tracked by requestId and status; the SAP object ID is returned on completion

Generic infrastructure versus business solution. The mapping example translates a Cognite InField observation into an SAP maintenance notification. The domain is plant maintenance observations from the field, and the service holds no decision logic. It is infrastructure that moves a payload from CDF into SAP.

Relevance to Hilti AMS and DI.

- It shows that governed, asynchronous writeback into SAP over OData is a mature pattern, and that a maintenance notification is the one repair-adjacent SAP object it handles.
- It requires CDF as the source system, so it is not a route for DI to write to SAP.
- It does not support the entities a repair, parts reuse or disposition workflow would write (service or repair orders, stock movements, reservations), so it provides no evidence that SAP writeback for those objects is solved.

Do not generalise. The scope applies to Cognite's service only. It does not describe what SAP can do natively, nor what any other integration product supports.

6. WSO2 SAP Sales Order connector: exact scope

The WSO2 connector is a third-party, open-source Ballerina client for one SAP OData API, Sales Order. It is neither an SAP-native adapter nor an aftermarket solution (class **E4**). Source: WSO2 SAP Sales Order connector overview, page shown as last updated 10 Oct 2026, read 5 Nov 2026.

Aspect	What the documentation states
Classification	WSO2 connector (package ballerinax/sap.s4hana.api_sales_order_srv), maintained as an open-source project in the Ballerina platform
SAP interface	SAP S/4HANA Sales Order OData (A2X) API, covering order processing from creation to delivery and billing
Operations	Create, read, update and delete for sales orders and sub-entities; OData query options ($filter, $select, $expand, $orderby, $top, $skip)
Sub-entities	Line items (quantity, material, plant), billing plans at header and item level, schedule lines, partners (sold-to, ship-to, bill-to), pricing elements, texts, related objects
Process flow	Navigation to preceding and subsequent document relationships
Setup	Configure S/4HANA Communication Management to expose the Sales Order (A2X) API and obtain credentials
SAP editions and versions	Not stated on the overview page

Relevance to Hilti AMS

- *Aftermarket spare-parts sales:* directly relevant to creating or reading a customer sales order for parts, including schedule lines and pricing. It would let an integration raise a parts order in SAP.
- *Parts procurement:* not covered. Procurement uses purchase orders and requisitions, which are different APIs.
- *Service sales and billing:* partly covered through billing plans on sales orders. Service contracts and service billing documents are not described.
- *Repair orders and service execution:* not covered. A sales order is not a repair or service order.
- *Component recovery and stock:* not covered. No stock, batch, serial-number or goods-movement operations appear.

What the connector does not establish. It does not show that SAP or WSO2 can decide a route, check eligibility, optimise allocation, hold approvals or reconcile outcomes. It also does not show how Hilti sells spare parts in SAP. It is a data-access client for one SAP API.

Implication. A sales-order connector illustrates that individual SAP APIs have ready-made clients. It is evidence about integration effort for one object, and not evidence that DI-like decision functionality exists off the shelf.

7. SAP aftermarket-service API and event matrix

Verification limit. The SAP Business Accelerator Hub (api.sap.com) blocks automated access, so API names and operations could not be confirmed on the hub itself. Verification levels: **V1** SAP Help or SAP Learning text read; **V2** SAP Community, SAP KBA or SAP developer material; **V3** third-party listing or the DI HTML only; **V4** not verified. An API in this table is not shown to be enabled in Hilti's environment. Edition is stated only where a source states it. In-house repair is shown by its current name, In-House Service, with the older name where relevant.

Area	API or event	Edition per source	Entity	Read or write	Protocol	Relevant DI workflow	Limitations and notes	Verification
Service orders	API_SERVICE_ORDER_SRV	Cloud (SAP Help); deprecated from release 2025 in Private Edition	Service order and items (A_ServiceOrder, A_ServiceOrderItem)	Create, read, update, delete; can also process repair orders	OData V2 (A2X)	Handoff of an approved repair or refurbish instruction; reading repair status	Key fields generated on create; deprecated in Private Edition 2025	V1 SAP Help
Service orders	OP_SERVICEORDER_0001 (service ServiceOrder, group API_SERVICEORDER)	Private Edition 2025 and later; on-premise help describes enhancements	Service order, partners, items, release for billing action (V1); tool equipment of items for serialised tools appears in an SAP Help excerpt only, as that page blocked access	Per V4 model; successor of the V2 API; a SOAP service-order API also exists (V1)	OData V4	As above, with serialised tool data	Model differs from V2, so V2 calls need adaptation	V1 SAP Learning, SAP Help
In-house service (formerly in-house repair)	In-House Service application (formerly In-House Repair; solution process 7GB in Private Edition and on-premise, scope item 3XK in Public Edition per SAP Learning); service orders via the service order API	Public Edition scope item 3XK; process in S/4HANA since 2020	In-house service objects (formerly repair objects), service quotations, service orders, service confirmations (older repair-specific transaction types deprecated from 2023 FPS02 per SAP Learning, V1)	Application functions; API coverage of in-house service objects not verified	Apps; OData where exposed	Evidence on repair status; instruction to start repair	Whether Hilti uses it is unknown	V1 for the process, V4 for in-house service object APIs: SAP Learning
Maintenance notifications	API_MAINTNOTIFICATION (OP_API_MAINTNOTIFICATION on the hub)	Not stated; Cognite supports it for S/4HANA on-premise 2021 FPS01+ and Cloud	A_MaintenanceNotification	POST and GET confirmed by an SAP KBA; Cognite shows create and update	OData V2	Raise or update an issue record tied to a tool or component	KBA reports slow responses on full-entity reads	V2 SAP KBA 3787252; Cognite doc
Service requests	Not identified	n/a	n/a	n/a	n/a	n/a	No service-request API identified in sources reviewed	V4

Area	API or event	Edition per source	Entity	Read or write	Protocol	Relevant DI workflow	Limitations and notes	Verification
Equipment and technical objects	Equipment, Functional Location, Measuring Point, Measurement Document APIs	Cloud, from release 2008	Equipment (tool master as technical object)	Create, update, read	OData	Asset identity and condition evidence	Release and edition coverage not re-verified	**V2** SAP Community, 2008
Product and material master	API_PRODUCT_SRV (A_Product)	Cloud per listing	Product (material)	Read; flat-header update only (deep updates not supported per KBA 2833338, as cited by a third-party listing)	OData **V2**	Compatibility and part identity evidence	Child entities (plant, valuation) via other endpoints	**V3** listing
Material stock	API_MATERIAL_STOCK_SRV (A_MatlStkInAcctMod)	Cloud per listing; used in the DI HTML	Stock quantities, composite key	Read	OData **V2**	Availability of recovered or new parts	Composite 11-field key; filter rather than key lookup	**V3** listing, DI HTML
Goods movements	API_MATERIAL_DOCUMENT_SRV	Cloud (SAP Help link on listing)	Material documents	Read; posting goods movements	OData **V2**	Record recovered-part stock moves, actual consumption	Posting rules and movement types are Hilti configuration	**V2** listing with SAP Help link
Reservations	API_RESERVATION_DOCUMENT_SRV	Not stated; used in the DI HTML	Reservation header	Create	OData **V2**	Reserve a recovered part for a repair	Not independently verified	**V3** DI HTML only
Batch and serial numbers	Not identified as APIs; serial numbers are consumed in repair or service confirmations (application feature)	Public Edition 2302	Serial numbers of stock service parts	Application feature	Apps	Component identity and traceability	API access to serial or batch data not verified	**V2** SAP Community, 2302
Bills of material	API_BILL_OF_MATERIAL_SRV	Not stated	BOM	Read (operations not verified)	OData **V2**	Component relationships and compatibility	Not verified	**V3** SAP Community thread
Quality inspection and notifications	Inspection Lot service (communication scenario SAP_COM_0318); quality notification API not identified	Cloud	Inspection lot	Per SDK package	OData	Quality evidence for reuse gates	Quality notification API not verified	**V2** SAP Cloud SDK package
Procurement	API_PURCHASEORDER_PROCESS_SRV; purchase requisition API API_PURCHASEREQ_PROCESS_SRV, deprecated since Public Edition 2402, successor API_PURCHASEREQUISITION_2 (OData **V4**)	Cloud per listing	Purchase order, purchase requisition	Create, read, update	OData **V2** and **V4**	Baseline of buying a new part; trigger purchase	Deprecation affects requisition integration	**V3** listing

Area	API or event	Edition per source	Entity	Read or write	Protocol	Relevant DI workflow	Limitations and notes	Verification
Spare-part sales	API_SALES_ORDER_SRV	Cloud per listing; WSO2 connector exists	Sales order, items, partners, pricing	CRUD	OData **V2** (A2X)	Parts sale demand	Not a repair or disposition decision interface	**V3** WSO2 connector page, listing
Billing and service costs	API_BILLING_DOCUMENT_SRV (read, cancel, get PDF); service order action ReleaseForBilling	Public Edition per community answer	Billing document	Read	OData **V2**	Actual proceeds and costs for reconciliation	Service cost objects not verified	**V2** SAP Community answer
Attachments and documents	API_CV_ATTACHMENT_SRV (OP_API_CV_ATTACHMENT_SRV_0001 on the hub)	Per Cognite	Attachments to SAP objects	Create	OData **V2**	Attach inspection evidence	Cognite uses it for notification attachments only	**V3** Cognite doc
Events	Maintenance notification due event sent to Event Mesh (calibration scenario); direct consumption of S/4HANA events in Advanced Event Mesh	Public Edition in the scenario	Events	Publish and subscribe	AMQP, Event Mesh	Trigger a case from a service or stock event	Event names for service orders and stock not verified	**V2** SAP Community, SAP deck

Reading. SAP exposes the objects a repair, parts and recovery workflow would touch: service and repair orders, notifications, equipment, stock, goods movements, procurement, sales and billing. It does not expose a recovery-route decision object. The weakest-verified areas are in-house service object APIs, serial and batch access, reservations, quality notifications and service-order events.

8. Combined BCG + SAP capability map

Position codes. A documented standard capability of Circelligence. **B** Hilti publicly confirmed using it. **C** SAP provides a relevant standard capability. **D** BCG + SAP could implement through configuration. **E** BCG + SAP could implement through custom development. **F** DI explicitly represents it in the HTML. **G** no clear DI differentiation. **H** uncertain or requires further evidence. "Not identified" means "not identified as a documented standard capability in the sources reviewed". DI entries describe the proposed product and refer to its HTML module names.

Reading. Rows 1 to 8 are Circelligence territory and DI has no differentiation there, which is expected because DI does not claim it. Rows 9 to 25 divide into capabilities where SAP supplies standard building blocks (approval, orchestration, writeback, agent governance) and capabilities where the sources reviewed identify no standard feature at all (route ranking, component matching, case-level economics, reconciliation). The second group is where any wedge must sit.

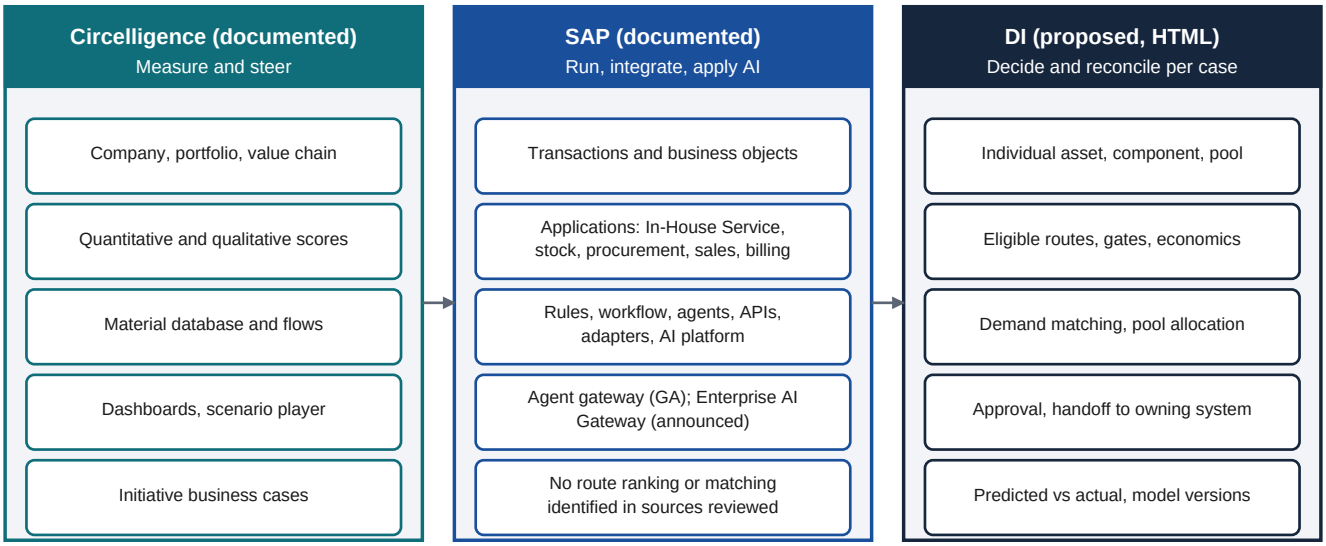
Capability	Circelligence current product	Confirmed Hilti deployment	SAP capabilities	Possible BCG + SAP implementation	DI capability in HTML	Position
Circularity scoring	Yes: quantitative and qualitative pillars	Score calculated (SAP 2023)	Hosts Circelligence on BTP	Standard	Sustainability view reports CO2e, not a circularity score	**A**, **B**, **G**
Quantitative material-flow analysis	Yes: full value chain from SAP data or Excel	Database of about 250,000 tons mapped (TechTarget)	SAP data ingestion	Standard	None	**A**, **B**, **G**
Qualitative assessments	Yes: survey-based readiness	Qualitative score cited (2023 report)	None	Standard	None	**A**, **B**, **G**
Material database and value-chain tracking	Yes	Automated database updates (SAP 2023)	BTP	Standard	None	**A**, **B**, **G**
Circularity dashboards	Yes	Dashboard UI (TechTarget)	Analytics services not reviewed	Standard	Command Center and Analytics are case and portfolio operational views	**A**, **B**, **G**
Strategic scenario modelling	Yes: scenario player	Not confirmed; 2023 BCG quote says groundwork was being laid for it	None	Standard	What-if on a single case in Route candidates, not strategy scenarios	**A**, **H**
Initiative prioritisation	Yes (scenario player)	Not confirmed	None	Standard	Ranked actions per case	**A**, **H**, **G**
Circular business-case assessment	Yes, for initiatives	Not confirmed	None	Standard	Per-case route economics, a different level	**A**, **H**
Repair-versus-replace decisions	Not identified	Customer repair-cost-limit policy; internal rule unknown	In-House Service quotations and prechecks; follow-up decisions on in-house service objects	E	No standalone module; repair cost is a route term	**C**, **E**, **H**
Returned-tool disposition	Not identified	Process confirmed (assess, pool, donate, dismantle, recycle); method unknown	Returns order and delivery as follow-up steps; stock and service APIs	E	Seven routes, six gates, Recovery waterfall	**C**, **E**, **F**
Component condition and eligibility	Not identified	Testing and criteria confirmed as Hilti's process (2022)	Inspection lot services; Build Process Automation rules	D, E	Six gates, evidence checklist, Condition and eligibility model	**C**, **D**, **E**, **F**

Capability	Circelligence current product	Confirmed Hilti deployment	SAP capabilities	Possible BCG + SAP implementation	DI capability in HTML	Position
Recovered component assessment	Not identified	Each part tested individually with new-part guarantee (2022)	As above	D, E	Evidence checklist, Evidence gaps	**B** (Hilti process), **C**, **F**
Recovered-component-to-dem and matching	Not identified	Not described	Service-part consumption, reservations and requisitions; no matching optimiser identified	E	Demand opportunity model, find_demand, Pool optimizer	**E**, **F**, **H**
Component and tool-level economic optimisation	Not identified (strategic economics only)	Part-type value versus CO2 analysis (TechTarget)	No optimiser identified	E	Route economics, cost trace	**E**, **F**, **H**
Multi-route recovery decisions	Not identified	Routes exist; choice method unknown	None identified	E	Seven routes with ranking	**E**, **F**, **H**
Pool and inventory optimisation	Not identified	Loan and Tools on Demand pools exist	Inventory functions; planning products not reviewed	E	Pool optimizer (exact allocation)	**E**, **F**, **H**
Service-centre decision workflows	Not identified	Automated criteria-guided extraction process (2022)	In-House Service; Build Process Automation	D, E	Ten stages, Case run	**B**, **C**, **G**, **H**
Deterministic eligibility gates	Not identified	Criteria exist; machine enforcement unknown	Build Process Automation rules	D	Six gates; no recommendation when a gate fails	**C**, **D**, **F**
Human approval	Not identified	Not described	Workflows; human-in-the-loop in agents	D	Approvals inbox, Approver delegations	**C**, **D**, **F**, **G**
Agent governance	Not identified	Not described	AI Agent Hub with runtime verification, AI Core tracing, agent gateway (GA), Enterprise AI Gateway (announced)	D	Agent registry, Permission matrix, write gate	**C**, **F**, **G**
Process orchestration	Not identified	Not described	Build Process Automation; Integration Suite	D	Process orchestration	**C**, **F**, **G**
Enterprise-system writeback	Not identified	Not described	S/4HANA APIs; Integration Suite adapters	D	Handoff; ERP, EAM, OMS connectors	**C**, **F**, **G**
Decision provenance and audit	Not identified	Per-tool repair history (2022)	Platform logging and tracing	E	Fetch log, evidence snapshot	**C**, **E**, **F**, **H**
Actual-outcome reconciliation	Not identified	Not described; data used to improve the process (2022)	None identified	E	Predicted vs actual, Overrides	**E**, **F**, **H**
Closed-loop decision learning	Not identified	Not described	AI Core (not reviewed for MLOps)	E	Model versions	**E**, **F**, **H**

9. DI versus BCG + SAP: what actually differs

DI and Circelligence answer different questions at different levels, so they are not substitutes. The overlap with SAP is real but sits in generic infrastructure. DI's proposed distinction is the case-level, domain-specific decision content that sits between the two, not generic workflow, AI or integration.

Figure 1. Levels of responsibility: Circelligence, SAP and DI



Teal and blue columns describe documented products. The navy column describes a proposed product with no validated results. The columns are layers of purpose, not a sequence of systems. SAP can also host decision logic built on its building blocks.

Dimension	Circelligence (documented)	SAP (documented)	DI (HTML, proposed)	Assessment
Unit of analysis	Company, product portfolio, value chain, material	Transaction and business object	Individual returned asset, component, pool or repair case	Distinct
Question answered	How circular are we, and which initiatives would raise the score	Record, process and integrate business transactions	Which eligible route should this tool or component take, and did it work	Distinct
Cadence	Periodic assessment (Hilti evaluates yearly)	Real-time transactions and events	Per case, event and approval driven	Distinct
Primary output	Circularity scores, dashboards, scenario results	Documents, APIs, workflows, agents	Ranked eligible routes with reasons, approval, instruction, reconciled outcome	Distinct
Economics	Business cases for improvement initiatives	Cost and valuation in ERP	Signed cost terms per route with uncertainty, cost trace	Different level; DI economics not shown to be validated
Sustainability metric	Circularity score from material flows	Not a standard decision input identified	CO2e avoided estimated versus measured	Different metric; no wedge
Eligibility and quality gating	Not identified	Rules and inspection objects available	Six gates with reasons	SAP can host gates; DI carries the structure
Approval and execution	Not identified	Workflows, agents with human-in-the-loop, APIs, adapters	Approvals inbox, delegations, handoff to systems	Overlaps with SAP; not a wedge
Agent governance	Not identified	AI Agent Hub, AI Core, MCP and A2A; agent gateway generally available; Enterprise AI Gateway announced	Agent registry, permissions, write gate, MCP client	Overlaps with SAP; weak

Dimension	Circelligence (documented)	SAP (documented)	DI (HTML, proposed)	Assessment
Recovered-part matching	Not identified	Service-part consumption, reservations and requisitions; no optimiser identified	Demand model, Pool optimizer	Candidate wedge, unproven
Outcome reconciliation and learning	Not identified	Not identified as a prebuilt decision function	Predicted vs actual, Model versions, Overrides	Candidate wedge, unproven
Domain content for Hilti	Hilti's materials and circularity data	Hilti's transactional data	None Hilti-specific	DI would need Hilti rules, classes and data

Three honest observations

1. DI is not Circelligence at operational level. Circelligence measures circularity of a portfolio. DI decides what to do with a specific item. A measure and a decision sit on different sides of the same data.
2. DI's approval, orchestration, writeback and agent-governance features overlap with generic SAP capabilities that are documented and current. They help DI operate but do not differentiate it.
3. DI is the only one of the three whose HTML describes a closed loop over case-level route decisions. That is the sole basis for a wedge, and the wedge depends on whether Hilti already has equivalent logic in its repair-centre process, a custom BTP build or an SAP application it already runs.

10. Candidate wedges and critical assessment

Classification scale: **Strong** = the DI HTML explicitly supports a valuable capability not identified as a standard documented feature in Circelligence or SAP. **Conditional** = value possible but needs data, development, integration or validation. **Weak** = SAP or Circelligence provides substantially equivalent standard functionality. **No credible wedge** = duplicates or adds no defensible increment. No candidate qualifies as Strong, because the DI claims are not validated and Hilti's own logic is unknown. Where a point is Uncertain, it is marked so.

Summary

Wedge	Classification	Main reason
A. Case-level aftermarket decision intelligence	Conditional	Explicit in the HTML; not identified in Circelligence or SAP; Hilti already runs an automated extraction process and may hold equivalent logic
B. Component recovery-to-demand matching	Conditional (highest unknown)	Explicit in the HTML; no SAP or Circelligence standard matcher identified; unknown whether Hilti already matches
C. Multi-route recovery optimisation	Conditional; folds into A	Seven routes ranked under constraints; no route optimiser identified in SAP; HTML omits donation and loaner routes
D. Governed decision-to-execution	Weak as a standalone claim	Every building block is a current SAP capability; only the domain-specific loop content is distinctive
E. Aftermarket economic optimisation	Conditional; folds into A and C	Case economics differ from Circelligence's initiative business cases, but the HTML's economics are structure, not validated optimisation

A. Operational, case-level aftermarket decision intelligence

Question	Assessment
1. In the DI HTML	Yes: Returned assets, Asset intelligence, Route candidates, Six eligibility gates, evidence checklist, cost trace. Not present: a repair-versus-replace module, Hilti service-history data model, remaining-life models
2. Circelligence	Not identified at case level; its documented scope is company, portfolio and value chain
3. SAP	In-House Service handles in-house service objects, prechecks, quotations, service orders and follow-up logistics; its precheck is a decision point on follow-up steps, and no route-ranking feature was identified
4. BCG + SAP could build	Yes, with rules, AI Core and workflows on BTP. Hilti may prefer this if it already owns the data and the automated repair-centre process
5. Hilti reason to build internally	Strong: it owns the value chain and describes an automated, criteria-guided extraction process (2022)
6. Frequent and material	Frequency: more than 1 million returned items a year (Hilti-reported); materiality of better route choice: unknown
7. Data and integrations	Condition and test evidence, repair history, cost, stock, demand, quality rules; ERP, repair, WMS, QMS, PLM sources; Hilti internal access **E7**
8. Operational result and measure	Net value per returned unit, reuse rate, parts avoided, compared with historical dispositions
9. Hard to reproduce	Structure no; calibrated content and outcome history yes, after use
10. Strongest evidence	HTML architecture plus Hilti's documented multi-route disposition and the absence of such a feature in the sources reviewed

B. Component-level recovery-to-demand matching

Question	Assessment
1. In the DI HTML	Yes: Demand opportunity model (confirmed versus forecast), find_demand, Pool optimizer, selective harvest with one destination per component, location and stock sources
2. Circelligence	Not identified
3. SAP	Spare-part consumption, serial numbers, reservations and requisitions for service parts exist in In-House Service and S/4HANA logistics; no matching optimiser identified; DI's reservation call is illustrative in the HTML
4. BCG + SAP could build	Yes, as custom development
5. Hilti reason to build internally	Moderate: parts reuse is large (190,000 in 2024) and extraction is automated; matching may already be part of that
6. Frequent and material	Frequent given parts volume; materiality unknown

Question	Assessment
7. Data	Recovered inventory by part and location, repair-order demand, compatibility, new-part lead time and price; integration with repair, parts and stock systems
8. Operational result and measure	Share of recovered parts consumed, new-part purchases avoided, stock ageing, repair wait
9. Hard to reproduce	Algorithm no; part-level demand and outcome data yes
10. Strongest evidence	Explicit HTML modelling; no equivalent in Circelligence or SAP documentation reviewed; Hilti's current matching is Unknown

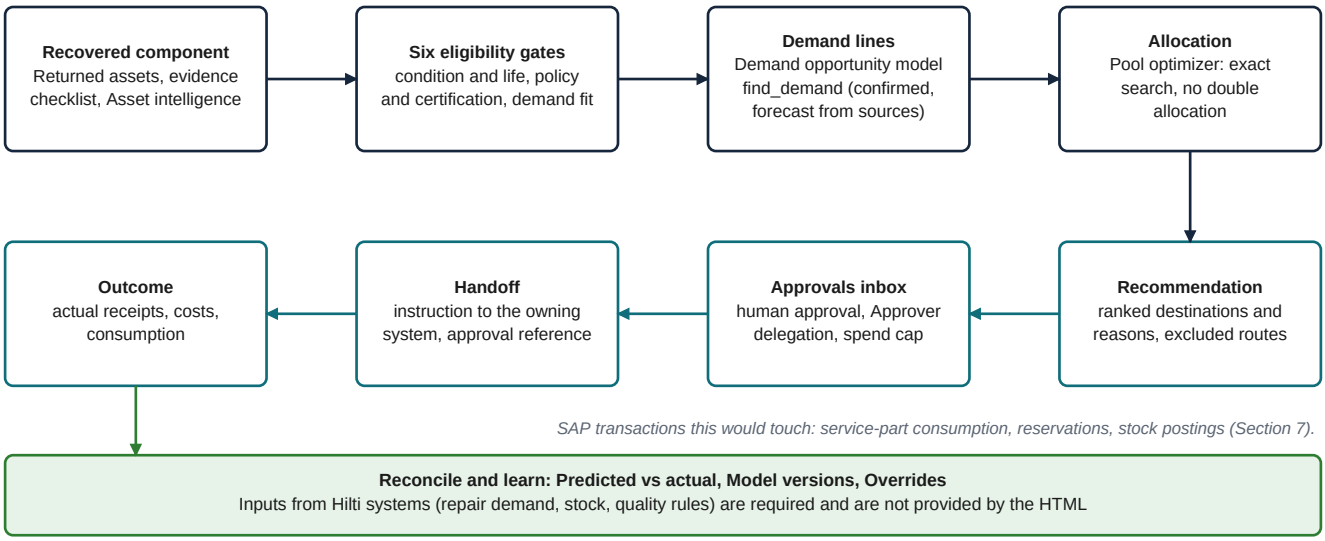
C. Multi-route operational optimisation. The HTML compares redeploy, resale, re-lease, internal parts, external parts, material recovery and responsible exit with gates and constraints. Hilti's routes are loan and On Demand pool, refurbish, donate, parts reuse and recycle. Overlap is partial: donation and loaner routes are absent from the HTML, and repair is a cost term rather than a route. Circelligence measures the circularity effect of such choices and does not make them. SAP provides no route optimiser in the sources reviewed. Classification: conditional, as a component of wedge A.

D. Governed decision-to-execution loop. Evidence collection, deterministic gates, human approval, permission-controlled agents, approved handoff and reconciliation all appear in the HTML. SAP documents each building block: Build Process Automation workflows and rules, Joule Studio agents with MCP and human-in-the-loop, AI Agent Hub governance, Integration Suite adapters and S/4HANA APIs, with an agent gateway generally available and an Enterprise AI Gateway announced (Section 3B). A specialised product could still offer the loop prebuilt for aftermarket, but the generic loop is not a wedge. Classification: weak as a standalone claim.

E. Aftermarket economic optimisation. Circelligence documents business cases for modelled initiatives, which is a strategic function. DI models net contribution per case with uncertainty and traces each cost line. That is a different level, but the HTML ranks routes by net value and allocates pools; it does not show labour scheduling, repair-capacity planning or a validated optimiser, and its baseline for uplift is modelled. Classification: conditional, merged into A and C.

Figure 3. Proposed DI workflow for recovered-component-to-demand matching

Proposed design from the DI HTML. No validated Hilti integration exists. Module names as in the HTML.



10A. What the DI HTML supports, stated precisely

Claim	What the HTML represents	What it does not represent	Wording to use
Repair-versus-replace	Route candidates with preparation or refurbishment cost terms, a "Refurbish, then sell" variant, the Condition and life gate, the Intervention model (process, predicted condition, cost, duration, success probability, capacity, required tests), and a predict_outcomes tool that lists repair cost and success among its predictions	A standalone repair-versus-replace function or output; a repair-cost model for Hilti tool families; customer quote logic	"Compares repair, refurbish and reuse routes against alternatives on route economics." Not "repair-versus-replace engine"

Claim	What the HTML represents	What it does not represent	Wording to use
Route economics versus a repair-cost model	Signed cost terms per route (avoided purchase, preparation and transfer, service exposure, residual value, allowance) with a cost trace by line; demo tables per asset class for labour, operations and logistics	A dedicated estimator of repair labour hours or parts for specific tools; the class tables are seeded illustrations	"Route-level economics with a cost build-up." Not "repair cost model"
Condition and eligibility versus inspection or certification	A Condition and eligibility model (inspection reference, grade, life assessment, qualification, certificates, policy version), a ten-item evidence checklist, six gates, and a tool that requests missing evidence	Performing inspection or tests, grading, or issuing certificates; these come from Inspection, MRO, QMS and Engineering sources	"Checks that Hilti's criteria were applied and evidenced." Not "assesses physical condition"
Matching versus forecasting or scheduling	Demand opportunity model with confirmed or forecast status taken from source systems, find_demand, and a Pool optimizer that allocates scarce units across demand lines without double allocation	A demand-forecasting model for repair parts; technician or repair scheduling; continuous re-matching. Time limits with escalation on execution steps are process clocks	"One-off allocation of recovered parts to known or expected demand." Not "demand forecasting" or "service scheduling"
Illustrative configuration versus validated integration	Source entries with vendor names, example-domain URLs, endpoints, authentication modes and "live" statuses, including SAP S/4HANA API names	Any connection to a real system; the entries are demonstration configuration	"Illustrates a connector configuration." Not "integrates with SAP"
Proposed capability versus demonstrated results	Seeded cases, model versions with improving error, predicted-versus-actual views and uplift figures	Any measured customer result	"Proposed design." Never quote HTML figures as performance

What DI could add beyond Hilti's automated extraction and testing process (Hypothesis). Hilti documents an automated, criteria-guided process that defines when each part is extracted and a test of each reused part (2022). Nothing here shows Hilti lacks case-level decision logic. If such logic exists, DI adds nothing there. If it covers extraction only, DI could add: a net-value comparison for the whole item before dismantling (pool reuse, refurbish, harvest or recycle); the destination and demand for each extracted part; stated reasons when a route is blocked; and a comparison of the outcome with the prediction. Each is **E7** for Hilti.

10B. Wedge reassessment after the corrections

Wedge	Previous	Now	Reason
A. Case-level aftermarket decision intelligence	Conditional	Conditional, narrower	SAP In-House Service has a precheck decision point but no ranking was identified; DI has no standalone repair-versus-replace function, so A is about returned-tool and component route decisions, not repair quotes
B. Recovered-component-to-demand matching	Conditional, highest unknown	Conditional; still the strongest candidate	Explicit in the HTML; SAP supports the underlying transactions but no matcher was identified in SAP or Circelligence; Hilti's own matching is unknown
C. Multi-route recovery optimisation	Conditional, folds into A	Unchanged	Seven routes in the HTML, but no donation or loaner route and no validated optimiser
D. Governed decision-to-execution	Weak	Weak, reinforced	Agent gateway is GA, AI Agent Hub has runtime verification, Joule Studio is GA, and the Enterprise AI Gateway is announced
E. Aftermarket economic optimisation	Conditional, folds into A and C	Unchanged	Route economics are a structure with uncertainty; not a repair-cost model and not a validated optimiser

No candidate is Strong. B remains the strongest candidate because it is the one element that is both explicit in the HTML and absent as a standard feature from the Circelligence and SAP documentation reviewed. That is a conditional position, not a proven one.

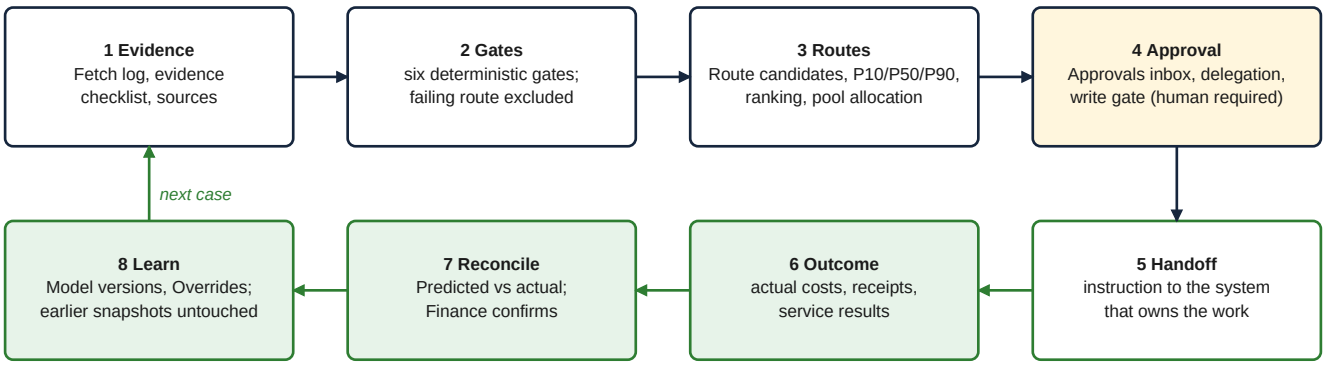
Wedge B in detail

Item	Assessment
DI HTML modules	Demand opportunity canonical model; find_demand and optimize_pool tools; Pool optimizer; Internal recovered-part use route and selective harvest (one destination per component, shared teardown cost once); Returned assets and Asset intelligence; Policy and certification gate; ERP and WMS sources in the Connection Fabric; Approvals inbox and handoff

Item	Assessment
SAP capabilities already supporting the underlying transactions	In-House Service confirmations consume service parts, including serialised parts; reservations and purchase requisitions for service parts; customer-stock and intracompany logistics for service objects; stock and goods-movement APIs (V3, Section 7)
Standard functionality not identified	A component-to-demand matcher or allocation optimiser in SAP or Circelligence, in the sources reviewed
What BCG and SAP could still build	A custom matching and allocation service on BTP using SAP APIs, rules and AI Core; the building blocks are documented
What Hilti may already do	Part extraction by automated criteria and individual part testing (2022); additions to its SAP landscape so reused items keep an identity and link to the parent tool (trade press, 5 Feb 2024); matching to repair demand may exist in TSC or parts systems (E7)
Data required	Recovered inventory by part, grade and location; repair-order demand by part and expected demand; compatibility; new-part price and lead time; transport cost; allocation and quality rules
Outcome that would show value	Share of recovered parts consumed within a set period, new-part purchases avoided, recovered-stock ageing and repair wait, compared with historical practice
Position	Conditional, not proven

Figure 4. Proposed DI decision-to-outcome feedback loop

Proposed design from the DI HTML. The loop closes when reconciled outcomes inform later decisions; improvement shown in the HTML is seeded.



Human approval is mandatory before any commitment of stock, spend or price. Retraining never overwrites an earlier evidence snapshot.

11. Recommended positioning, defensibility and moat

Primary wedge (conditional): a case-level aftermarket decision layer that, for each returned tool and recovered component, determines which routes are eligible under Hilti's own rules, ranks them on case economics with stated uncertainty, matches recovered components to open repair demand, routes the choice for approval, hands the instruction to Hilti's systems, and reconciles the actual result. Recovered-component-to-demand matching is the sharpest element, and outcome reconciliation is the proof mechanism.

Positioning statement. DI is a proposed domain decision layer for returned tools and recovered parts. Using the facts held in SAP and other systems and the rules Hilti owns, it would rank eligible recovery routes and part allocations with stated economics and uncertainty, route the choice for approval into the system that executes it, and compare the outcome with the prediction. Circelligence measures circularity at portfolio level. SAP supplies the applications, rules, workflow and AI on which the transactions run, and on which a comparable layer could also be built.

Why the statement was revised. The earlier wording described SAP as what "records and executes what happens". That oversimplifies SAP. It also provides a repair application with a precheck decision step (In-House Service), business rules, workflow, human-in-the-loop agents and an AI platform that SAP now states is generally available. The distinction that survives scrutiny rests on documented domain content, not on architecture layers. Circelligence documents portfolio-level measurement and initiative scenarios. SAP documents transaction applications and general decision and AI tooling. In the sources reviewed, neither documents recovery-route ranking, component eligibility gates for recovered tools, recovered-part-to-demand matching or reconciliation of route decisions to outcomes. DI's HTML proposes that content. That is the whole basis of the positioning, and it holds only until Hilti shows equivalent logic inside its repair-centre process, a custom BTP build or an SAP application it already runs.

The HTML does support this proposition as an intended design: Returned assets, Route candidates, Six eligibility gates, Demand opportunity model, Pool optimizer, Approvals inbox, handoff and Predicted vs actual all exist as described modules. Support in the HTML is support for the design, not for its performance.

What this is, and is not. It is a publicly identifiable product gap: the sources reviewed identify no standard Circelligence or SAP feature for case-level route ranking, component-to-demand matching or decision-outcome reconciliation. It is not a claim that BCG, SAP or Hilti cannot build these. SAP documents the building blocks and a build on BTP is credible.

Why Circelligence does not obviously remove the need. Its documented scope is assessment, dashboards and initiative scenarios at company and value-chain level. It does not describe case-level decisions, eligibility gates, approvals, writeback or reconciliation. Its output is an input to strategy that DI-level decisions could be measured against.

Why SAP's platform does not automatically make DI redundant. SAP supplies workflow, rules, agents and integration as general-purpose building blocks, plus In-House Service as a domain application for repair intake, prechecks, quotations, service orders, confirmations and returns logistics. None of the sources reviewed describes SAP-supplied domain content for recovery routes, component eligibility, matching or reconciliation. A customer must either build and maintain that content or adopt a product that carries it. That is a build-versus-buy decision that SAP's capabilities leave open.

Why a customer might prefer a specialised product

- It arrives with a worked domain model (route candidates, gates, demand, outcome) rather than a blank platform.
- The decision, approval, handoff and reconciliation loop is already connected and documented in one place.
- Evidence provenance and decision snapshots are part of the design and are not bespoke work.
- It does not require Hilti to commit BTP project capacity to a non-core build.

These are arguments for a buying decision. They are not proof that DI is better than a Hilti-built solution.

Modules that support it: Returned assets, Asset intelligence, Route candidates, Six eligibility gates, Pool optimizer, Demand opportunity and canonical models, Connection Fabric, Approvals inbox and Approver delegations, Process orchestration, Predicted vs actual, Model versions, Audit and provenance.

Hilti systems and data required (E7 for access): repair and service records, tool and component identity and condition, inspection and test evidence, quality and reuse rules, parts stock and location, repair-order demand, cost and labour data, and S/4HANA service, stock and procurement objects where Hilti uses them.

What would make the wedge defensible

- Hilti-calibrated route economics and gate content that perform on real data.

- Reconciled decision-outcome history that accumulates with use.
- Part-level demand and outcome data that cannot be recreated without operation.
- Embedded approval and handoff workflows that carry switching cost.

Why no moat is currently proven. A moat is not established. It could only arise from these after use, and ownership of the learning is itself an open question. The model-error improvement shown in the HTML is seeded.

What must not be claimed

- That DI replaces or competes with Circelligence on circularity scoring, scenarios or material flows.
- That SAP or BCG cannot build this.
- That agents, MCP, workflow, integration or governance differentiate DI against SAP.
- That Hilti lacks case-level decision logic or part matching, which is unknown.
- That DI is integrated with, or has validated access to, any Hilti or SAP system.
- That any Hilti result (parts reused, tons saved) is DI's, or that any DI performance figure is measured.

12. Hilti objections and recommended answers

The answers accept that Hilti works with BCG and SAP and that both are capable. Where DI adds nothing, the answer says so. DI functionality is named as in the HTML (proposed). Evidence points to sections of this report and the sources cited in them.

Questions 1 to 10

Q1. We already use Circelligence. Why do we need DI?	
Concise answer	Circelligence tells you how circular Hilti is. DI would decide what happens to a specific returned tool or part and check the result. They answer different questions, so DI is a candidate complement, not a substitute.
Detailed answer	Circelligence documents scores, a material database, a dashboard and a scenario player at company and value-chain level. DI's HTML works per case: eligible routes, economics, matching, approval, handoff and reconciliation. Whether Hilti needs that depends on how it makes those choices today.
Supporting DI functionality (HTML)	Returned assets; Route candidates; Six eligibility gates; Predicted vs actual
Circelligence or SAP functionality	Circelligence pillars (BCG page); Hilti score and database confirmed (SAP, 2023)
Evidence	Sections 1, 2, 9
Claims to avoid	That DI measures circularity better or replaces Circelligence

Q2. Isn't Circelligence already tracking circularity across our product value chain?	
Concise answer	Yes, at portfolio and material level. DI does not repeat that; it works on individual items once they are returned.
Detailed answer	Hilti evaluates its portfolio yearly and uses the result for Scope 3 goals. Circelligence's documented data runs from raw materials to end of life and does not describe case-level decisions. DI's outputs could be measured against those scores but do not substitute for them.
Supporting DI functionality (HTML)	Returned assets; Route mix; Sustainability view
Circelligence or SAP functionality	Quantitative pillar; confirmed yearly portfolio assessment
Evidence	Sections 1, 2
Claims to avoid	That DI also tracks value-chain circularity

Q3. Circelligence already provides scenario analysis and business cases. What is different in DI?	
Concise answer	Circelligence's scenarios test improvement initiatives. DI values routes for one item with uncertainty and later compares the actual result.
Detailed answer	The scenario player models changes to processes and raw materials and supports business cases for initiatives. DI carries signed cost terms per route, P10 to P90 ranges, a what-if on a case, and Predicted vs actual. One sets priorities and the other applies them item by item. DI's economics are not validated.
Supporting DI functionality (HTML)	Route candidates; cost trace; Predicted vs actual
Circelligence or SAP functionality	Scenario player; initiative business cases
Evidence	Sections 1, 8, 9
Claims to avoid	That Circelligence cannot calculate economics

Q4. We already recover and reuse spare parts. What problem is DI solving?	
Concise answer	Not recovery itself. Possibly which parts to recover and where each goes, if that is not already optimised, which public sources cannot tell us.
Detailed answer	Hilti reused 190,000 parts in 2024 and runs an automated process that defines when each part is extracted using prescribed criteria (2022). DI would add net-value ranking, matching to repair demand and reconciliation. If that already exists in your process, the increment is small.
Supporting DI functionality (HTML)	Internal recovered-part use route; Demand opportunity model; Pool optimizer
Circelligence or SAP functionality	Hilti's repair-centre process; Circelligence tracks parts reuse as a metric
Evidence	Section 2
Claims to avoid	That reuse or criteria-based extraction is missing

Q5. We already have SAP BTP. Why introduce another platform?	
Concise answer	You may not need one. The question is whether to build and maintain a domain decision loop on BTP or adopt a product that carries it. DI does not replace BTP.
Detailed answer	SAP documents workflow, rules, agents, integration and APIs. None is described as providing route, eligibility, matching or reconciliation content. Building that content is the cost. DI would sit beside BTP and use SAP interfaces.
Supporting DI functionality (HTML)	Connection Fabric; Process orchestration
Circelligence or SAP functionality	Build Process Automation, Joule Studio, Integration Suite (Section 3)
Evidence	Sections 3, 9
Claims to avoid	That BTP cannot do this

Q6. Why can't we build the DI features directly in SAP?	
Concise answer	You can. The case for DI is time and ownership of domain content, not capability.
Detailed answer	Every building block is documented: rules, workflows, AI Core, agents with human oversight, adapters, APIs. DI offers a defined domain model and loop. Hilti decides on effort, maintenance and control. DI's advantage is unproven.
Supporting DI functionality (HTML)	Whole proposed loop
Circelligence or SAP functionality	As in Q5
Evidence	Sections 3, 11
Claims to avoid	That SAP development is hard or slow

Q7. SAP Integration Suite already connects our applications. Why do we need DI?	
Concise answer	It moves data between systems; it does not decide. DI could use it as the transport.
Detailed answer	The adapters cover OData, SOAP, IDoc, RFC, SFTP, AMQP, Kafka and Open Connectors. DI's Connection Fabric overlaps in connecting systems; its added layer is canonical models, a conflict queue, evidence, gates and ranking.
Supporting DI functionality (HTML)	Connection Fabric; Conflict queue; canonical models
Circelligence or SAP functionality	Cloud Integration adapters; Event Mesh
Evidence	Sections 3, 4
Claims to avoid	That DI's connectors beat SAP's

Q8. SAP supports AI agents, workflows and API orchestration. What is DI's advantage?	
Concise answer	On the generic side, little. The advantage would be prebuilt aftermarket decision content, not agents or orchestration.
Detailed answer	Joule Studio (the original Agent Builder is described as generally available since January 2026, and SAP's CTO is reported on 2 Nov as calling Joule Studio generally available) supports MCP and human-in-the-loop; AI Agent Hub governs agents and MCP servers with runtime verification; an agent gateway is generally available and an Enterprise AI Gateway has been announced. DI's registry, permissions and write gate are comparable in intent. DI's tools such as find_demand and optimize_pool are the domain content.
Supporting DI functionality (HTML)	Agent registry; Permission matrix; tool catalogue
Circelligence or SAP functionality	Joule Studio; AI Agent Hub; AI Core
Evidence	Section 3, 3B
Claims to avoid	That agent governance differentiates DI

Q9. Can Circelligence and SAP already provide the same decision functionality?	
Concise answer	Not as documented standard features in the sources reviewed. SAP could be used to build them.
Detailed answer	Circelligence is not documented at case level. SAP offers In-House Service (formerly In-House Repair) with a precheck decision on follow-up steps for in-house service objects, but no route-ranking or matching feature was identified. Hilti's private custom work cannot be seen.
Supporting DI functionality (HTML)	Route candidates; Pool optimizer; Predicted vs actual
Circelligence or SAP functionality	In-House Service; Circelligence scope
Evidence	Sections 8, 10
Claims to avoid	That Hilti lacks such logic

Q10. Is DI simply duplicating Circelligence at an operational level?	
Concise answer	No. Circelligence measures and DI decides. The only overlap is sustainability reporting, and the metrics differ.
Detailed answer	DI's Sustainability view reports estimated versus measured CO2e avoided by route. Circelligence scores circularity from material flows. Definitions would need alignment.
Supporting DI functionality (HTML)	Sustainability view
Circelligence or SAP functionality	Circelligence scoring
Evidence	Sections 8, 9
Claims to avoid	Treating CO2e as a circularity score

Questions 11 to 20

Q11. How is DI different from a custom BTP application?	
Concise answer	Functionally it could be the same thing. The difference is that it exists as a defined product rather than a project.
Detailed answer	A custom app needs the same modules. The trade-off is fit to Hilti and control against product maturity and carrying cost. Neither route is proven: DI has no validated results and a custom build would be new too.
Supporting DI functionality (HTML)	Whole proposed loop
Circelligence or SAP functionality	BTP application services (Section 3)
Evidence	Sections 3, 11
Claims to avoid	That DI is better than a custom build

Q12. Which exact SAP APIs would DI consume and which would it update?	
Concise answer	The HTML illustrates reads from material stock, purchase order and product APIs and writes to reservation and material document APIs. Hilti's actual set depends on its edition and processes and has not been verified.
Detailed answer	The DI HTML ERP source shows API_MATERIAL_STOCK_SRV, API_PURCHASEORDER_PROCESS_SRV and API_PRODUCT_SRV for reads, and API_RESERVATION_DOCUMENT_SRV and API_MATERIAL_DOCUMENT_SRV for writes. For repair, the service order API (API_SERVICE_ORDER_SRV, or OP_SERVICEORDER_0001 in Private Edition 2025 and later) and the maintenance notification API are relevant. The Business Accelerator Hub could not be read, so see the verification levels in Section 7.
Supporting DI functionality (HTML)	Connection Fabric ERP and EAM sources
Circelligence or SAP functionality	S/4HANA APIs (Section 7)
Evidence	Sections 4, 7
Claims to avoid	That these APIs are enabled at Hilti

Q13. Does DI require us to replace any existing SAP or BCG system?	
Concise answer	No.
Detailed answer	SAP stays the system of record and Circelligence stays the measurement tool. DI reads evidence and sends approved instructions. It needs read access and agreed write interfaces.
Supporting DI functionality (HTML)	Connection Fabric; handoff model
Circelligence or SAP functionality	Existing Hilti systems
Evidence	Sections 3, 9
Claims to avoid	That DI works without access to Hilti systems

Q14. What information can DI provide that our existing systems do not?	
Concise answer	A ranked, gated and explained recommendation for each case and its reconciled outcome, if those are not already produced elsewhere.
Detailed answer	Evidence per case with source and freshness, excluded routes with the failing gate, cost lines, an evidence snapshot with model and policy version, and predicted versus actual. Existing systems hold the facts; DI's output is the decision record.
Supporting DI functionality (HTML)	Route candidates; Fetch log; Predicted vs actual
Circelligence or SAP functionality	Facts in SAP; scores in Circelligence
Evidence	Sections 8, 9
Claims to avoid	"Single source of truth"

Q15. How does DI make a better repair-versus-replace decision?	
Concise answer	DI has no standalone repair-versus-replace function. It compares repair-and-reuse routes against retirement for Fleet and end-of-contract items.
Detailed answer	Repair or preparation cost, service exposure, remaining life from engineering sources, and alternatives are compared after gates. Hilti's Repair Cost Limit remains a commercial policy. SAP In-House Service covers prechecks, service quotations and service orders.
Supporting DI functionality (HTML)	Route candidates; Condition and life gate
Circelligence or SAP functionality	In-House Service; Hilti policy
Evidence	Sections 8, 10, 10A
Claims to avoid	That DI is a repair-versus-replace engine

Q16. How does DI determine whether a recovered component can safely be reused?	
Concise answer	It does not judge physical safety. It enforces Hilti's rules: missing evidence or a failing gate means no recommendation.
Detailed answer	Gates cover condition and life, post-intervention condition, and policy and certification, reading QMS, PLM and engineering sources. Evidence gaps are listed. Hilti tests each reused part individually (2022); those tests remain Hilti's.
Supporting DI functionality (HTML)	Six eligibility gates; Evidence gaps
Circelligence or SAP functionality	SAP inspection lot services; Hilti testing
Evidence	Sections 7, 8, 10A
Claims to avoid	That DI certifies parts

Q17. How does it match available recovered components to repair demand?	
Concise answer	It treats repair demand as demand lines, checks compatibility and eligibility, and allocates scarce parts so none is committed twice. It needs repair-demand and stock data.
Detailed answer	The Demand opportunity model carries specification, quantity, location, need-by and confirmed or forecast status. The Pool optimizer compares exact and greedy allocation. SAP supports reservations for service parts but no matcher was identified. The HTML models one-off allocation, not continuous scheduling or forecasting.
Supporting DI functionality (HTML)	Demand opportunity model; find_demand; Pool optimizer
Circelligence or SAP functionality	Reservations, requisitions and service-part consumption in In-House Service
Evidence	Sections 8, 10, 10A
Claims to avoid	Forecasting or scheduling claims

Q18. How does DI prevent an AI agent from executing an unauthorised action?	
Concise answer	Write tools that commit stock, spend or price stop at human approval, and agents cannot waive a failed gate.
Detailed answer	The Handoff Coordinator sends only what a person approved, capped at the approved spend; send_instruction needs an approval reference; every call is logged. SAP offers comparable controls (human-in-the-loop, AI Agent Hub), so the claim is that DI's are predefined for these actions.
Supporting DI functionality (HTML)	Agent registry; write gate; Call log
Circelligence or SAP functionality	Joule Studio human oversight; AI Agent Hub
Evidence	Section 3B
Claims to avoid	That DI's controls are safer than SAP's

Q19. How does DI demonstrate that its decisions improve actual service outcomes?	
Concise answer	By comparing predictions to actuals line by line and tracking model versions and overrides, once Hilti data flows. The figures in the HTML are seeded.
Detailed answer	Predicted vs actual, Model versions and Overrides are the mechanism. Uplift is measured against a modelled baseline, so Hilti's real baseline is required. A comparison with historical dispositions is the evidence that would matter.
Supporting DI functionality (HTML)	Predicted vs actual; Model versions; Overrides
Circelligence or SAP functionality	None identified as a prebuilt function
Evidence	Sections 8, 10
Claims to avoid	Quoting HTML improvement as a result

Q20. What is the genuine product moat if SAP can build similar functionality?	
Concise answer	None is proven. A possible one is reconciled outcome history combined with Hilti-tuned rules, and it exists only after use.
Detailed answer	Features are replicable. Data accumulates only with operation, and the outcome history created at Hilti may belong largely to Hilti.
Supporting DI functionality (HTML)	Model versions; Audit and provenance
Circelligence or SAP functionality	Platform building blocks
Evidence	Sections 10, 11
Claims to avoid	That a moat exists

13. Key uncertainties and unverified claims

Item	Status	Why it matters
SAP Business Accelerator Hub API details	Could not be read (the site disallows automated access). API names and operations come from SAP Help, SAP Learning, SAP KBA, community posts, a cloud SDK package page and third-party listings, graded **V1** to **V3** in Section 7	Exact API names, versions, operations and editions must be checked on the hub before any integration claim
SAP Help pages named for In-House Service	Three of four requested help.sap.com pages blocked automated access; the SAP Learning page was read in full; other facts rest on excerpts	In-House Service API coverage and Public Edition naming are unverified
"OData API: In-House Service"	Appears as a title in SAP Help what's-new navigation for 2025 FPS02; name, scope and operations not read	Do not describe in-house service object APIs as confirmed
In-House Service 2.0 transaction types	From a member-authored SAP Community post, not SAP official	Treat REPA, REPO, REPC to IHRP, SRVO, SRVC as unconfirmed
Hilti's SAP edition, modules and enabled APIs	Unknown	Decides which APIs exist for Hilti and whether S/4HANA Service or In-House Service (formerly In-House Repair) is used
Hilti's SAP BTP, Integration Suite, Build Process Automation and AI licences	Unknown	Decides whether SAP's building blocks are already available to Hilti
Whether Hilti uses Circelligence's scenario player and qualitative survey	Not confirmed; the 2023 BCG quote suggests scenario assessments were being integrated	Decides how much of Circelligence's strategic functionality is deployed
Private or customised Circelligence functionality at Hilti	Unknown	Public material is limited and may understate private work
BCG's 120,000 parts figure for 2022	Not matched by Hilti's 2022 report (100,000); Hilti's undated page says "up to 120,000"	Reporting period and basis of the figure are unclear
The "90 tons" figure	Reported unchanged beside 100,000, 120,000 and 190,000 parts	Cannot be a consistent per-year measure
Whether Circelligence caused Hilti's reuse results	BCG claim, untested	Results belong to Hilti and are not DI's
Cognite writeback beyond maintenance notifications and attachments	Not stated; setup page describes configurable endpoints but lists two supported entities. The API reference pages for endpoints were not opened	Whether other SAP entities can be used is **E7**
WSO2 connector's SAP edition and version support	Not stated on the overview page	Edition coverage is unknown
SAP Datasphere, HANA Cloud, Build Apps, Analytics Cloud, planning products	Not reviewed in detail	Optimisation, analytics and data capabilities in SAP are **E7** for these
Integration Suite monitoring and error handling specifics	Treated as a general product feature and not verified page by page	Relevant to the reconciliation and audit rows in Section 3
SAP Enterprise AI Gateway availability	**Announced** at TechEd on 28 Oct 2026 without a stated availability date; the agent gateway is stated as generally available	Do not describe the Enterprise AI Gateway as generally available
SAP Business AI Platform general availability	Reported statement by SAP's CTO on 2 Nov 2026 through SAP's Africa News Center; the November SAP News Center page named in the request returned an error	The status is an executive statement as reported
Hilti's current route-choice method, part matching and extraction criteria	Unknown beyond a 2022 description	The central uncertainty for every wedge
DI performance	None demonstrated; the HTML uses seeded data	No uplift, accuracy or learning claim can be made
DI's SAP API list	Illustrative configuration in the HTML, not validated against any landscape	Do not treat as a statement of what DI integrates with
Unverified integrations	No integration between DI and any Hilti, SAP or BCG system exists or is evidenced	Do not describe any as operational

Internal consistency flags raised by this pass

- The Open Connectors count differs between SAP sources (over 150 in the Feature Scope Description, 250+ connectors in a user-group deck). Both are reported; the difference is not reconciled.
- The statement that Event Mesh is included in Integration Suite standard and premium editions comes from a March 2025 SAP user-group deck and was not re-checked against the August 2026 Feature Scope Description excerpt.
- The Hilti parts-reuse figures in Section 2 come from pages with different dates, scopes and bases and are not a single time series. The Hilti New Zealand page reports global 2024 figures.
- Some SAP Learning and SAP Community sources cited for service functionality use the older "In-House Repair" name because they predate the rename; they are cited as the evidence they are, and the report uses In-House Service as the current term.
- The SAP Learning In-House Service lesson carries a course-version tag suggesting a 2026 course, but no page date.
- A member-authored SAP Community post places In-House Service 2.0 at 2025 FPS02, whereas SAP Learning says it was introduced with 2023 FPS02 and that the rename came with release 2025. The report follows SAP Learning.
- SAP's Q2 2026 highlights list the new Joule Studio under Early Adopter Care, while an SAP Community post and a reported CTO statement call Joule Studio generally available. Section 3B shows both.
- The Hilti page with "up to 120,000 parts" carries 2020 collection data, and TechTarget is dated 5 Feb 2024 (earlier text had called it undated).
- The framework details in Section 1A (indicator counts, grades, data effort, delivery model, maturity rating) come from the 2021 BCG and Circle Economy paper and may have changed in later product versions.

Source verification summary. Links are inline next to each claim. This pass tried to open every source that had previously been seen only as a search excerpt.

Status	Sources
Opened in full	BCG and Circle Economy white paper on the CIRCelligence indicators framework (Nov 2021); BCG Circelligence product page; SAP News Center Hilti announcement (22 Nov 2023); Hilti Group media release (22 Nov 2023); TechTarget (5 Feb 2024); Hilti 2021 and 2022 reports; Hilti New Zealand, Israel and US circularity or sustainability pages; Cognite writeback and setup pages; WSO2 Sales Order connector page; SAP Learning pages on in-house service, its configuration, and service-order APIs; SAP Q2 2026 Business AI highlights; SAP TechEd 2026 article (28 Oct); SAP Africa News Center article (2 Nov)
Blocked by the site (excerpt only)	SAP Business Accelerator Hub; all SAP Help pages and SAP Help PDFs, including the Integration Suite Feature Scope Description; SAP Community posts and threads; SAP Knowledge Base article
Not re-opened in this pass (excerpt only)	Hilti 2023 Sustainability Report (flipbook); SAPinsider TechEd article; SAP Architecture Center agent guide; SAP user-group Integration Suite decks (2025, 2026); SAP Cloud SDK release notes and package page; third-party API listings (Sim, API Evangelist); SAP Learning Public Edition in-house repair pages
Failed	The November SAP News Center page named in the request (a client error); its content was read from the SAP Africa News Center version